

WEEK 4 – ASSIGNMENT

Objective

The objective of this assignment was to understand the fundamentals of Microsoft Azure Cloud and build a complete end-to-end data pipeline using Azure Storage Account and Azure Data Factory (ADF). The pipeline validates a source CSV file using the Get Metadata activity and copies it from a source Blob container to a destination Blob container. This assignment also covers Azure RBAC (Role-Based Access Control) to securely connect Azure services.

Task 1: Azure Portal Exploration & Resource Group Creation

Objective

Create a Resource Group to organize all Azure resources required for the project.

Steps Performed

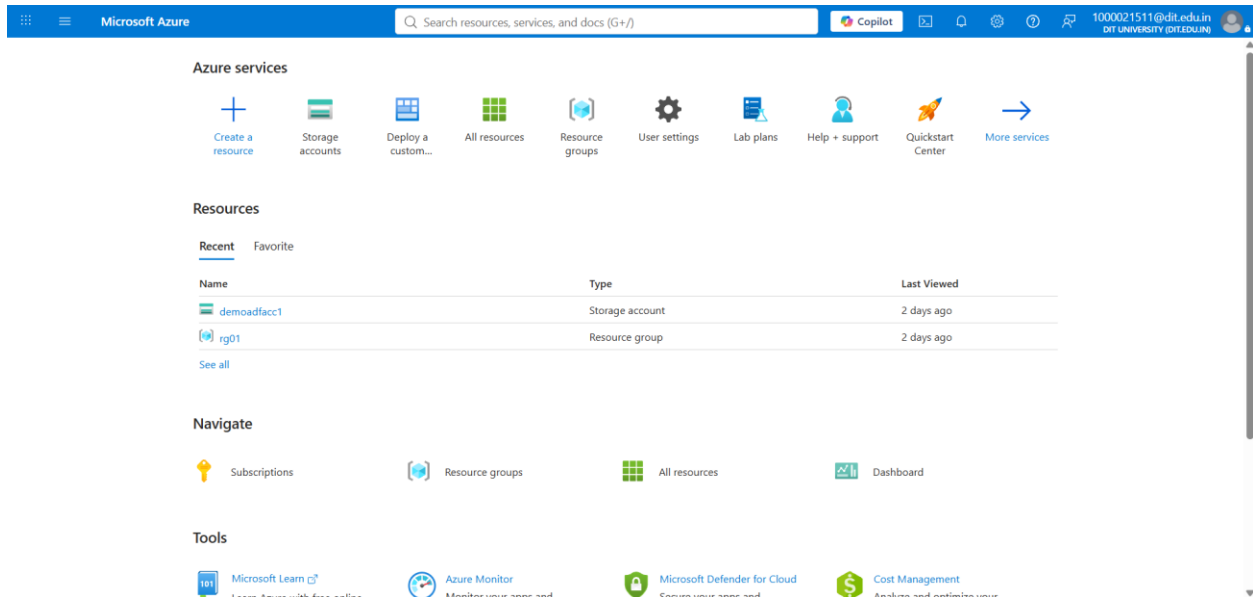
- Logged into the Microsoft Azure Portal using the Azure Student subscription.
- Explored the Azure Portal dashboard.
- Created a new Resource Group named **rg-celebal-adf-palak**.
- Selected the appropriate Azure region supported by the subscription.
- Verified that the Resource Group was created successfully.

Result

The Resource Group was successfully created and used to organize all Azure resources for the assignment.

Screenshot

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Task 2: Azure Storage Setup

Objective

Create Azure Blob Storage and upload the dataset.

Steps Performed

- Created a Storage Account named **stpalakadf2026**.
- Selected Standard Performance and Locally Redundant Storage (LRS).
- Opened the Storage Account after deployment.
- Created a Blob Container named **source**.
- Uploaded the **Sample-Superstore.csv** dataset into the container.

Result

Azure Blob Storage was successfully configured and the dataset was uploaded successfully.

Screenshots

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The screenshot displays three distinct sections of the Azure portal interface:

- Sample - Superstore.csv**: A blob properties page showing details for a CSV file. The URL is `https://stpalakad2026...`. Properties include: LAST MODIFIED (7/11/2026, 9:10:07 PM), CREATION TIME (7/11/2026, 9:10:07 PM), TYPE (Block blob), SIZE (2.18 MiB), ACCESS TIER (Hot (Inferred)), and SERVER ENCRYPTED (true).
- superstore-data**: A container overview page. It shows the authentication method as 'Access key' and a search bar. The table below indicates 'No items found'.
- Deployment completion**: A confirmation screen stating 'Your deployment is complete'. It provides deployment details such as the name 'stpalakad2026_1783784037291', subscription 'Azure for Students', and resource group 'rg-celebal-adf-palak'. It also includes a 'Go to resource' button and a sidebar with recommendations for cost management, Microsoft Defender for Cloud, and free tutorials.

Name	Last modified	Access tier	Blob type	Size	Lease state
No items found					

Property	Value
Deployment name	stpalakad2026_1783784037291
Subscription	Azure for Students
Resource group	rg-celebal-adf-palak

Task 3: Azure Data Factory Configuration

Objective

Configure Azure Data Factory and establish connectivity with Azure Blob Storage.

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Steps Performed

- Created a new Azure Data Factory instance named **adfpalak2026**.
- Opened the ADF Studio interface.
- Explored the Author, Monitor, and Manage sections.
- Created a Linked Service connecting Azure Data Factory with Azure Blob Storage.
- Created the Source Dataset for the uploaded CSV file.
- Created the Destination Dataset for the output container.

Result

Azure Data Factory was successfully configured and connected to Azure Storage.

Screenshots

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DS_Source_Superstore • DS_Destination_Sup...

DelimitedText
DS_Destination_Superstore

Properties

General Related

Name *
DS_Destination_Superstore

Description

Annotations
+ New

Cancel

Connection Schema Parameters

Linked service *
LS_BlobStorage

Test connection Edit + New Learn more

File path *
superstore-data / Directory / Superstore_Output.csv

Compression type
No compression

Column delimiter
Comma (,)

Row delimiter
Default (\r\n, or \n)

Encoding
Default(UTF-8)

Quote character
Double quote (")

Escape character
Backslash (\)

DS_Source_Supersto...

DelimitedText
DS_Source_Superstore

Properties

General Related

Name *
DS_Source_Superstore

Description

Annotations
+ New

Cancel

Connection Schema Parameters

Linked service *
LS_BlobStorage

Test connection Edit + New Learn more

File path *
superstore-data / Directory / Sample - Superstore.csv

Compression type
No compression

Column delimiter
Comma (,)

Row delimiter
Default (\r\n, or \n)

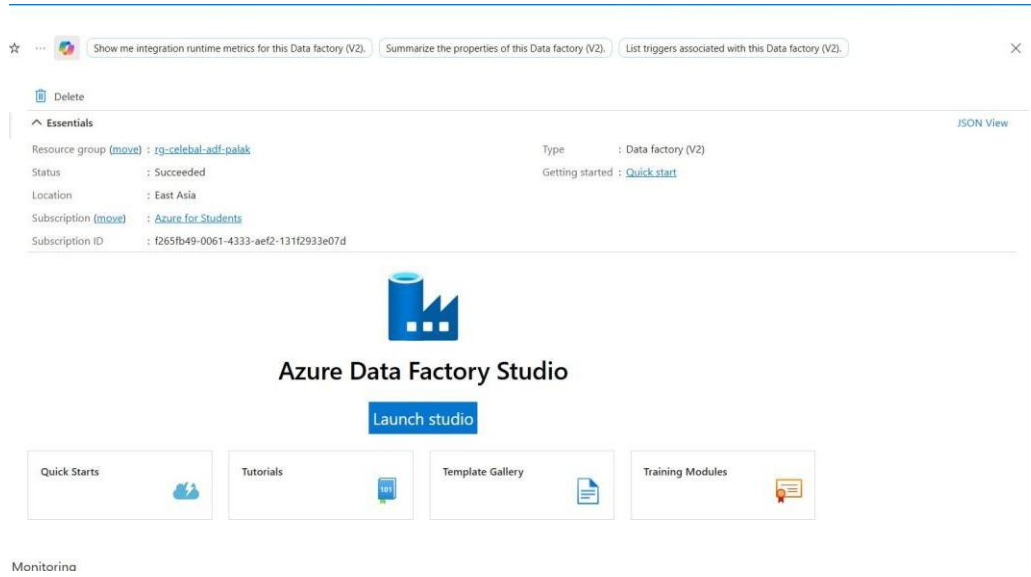
Encoding
Default(UTF-8)

Quote character
Double quote (")

Escape character
Backslash (\)

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Task 4: Pipeline Development

Objective

Build an end-to-end Azure Data Factory pipeline.

Steps Performed

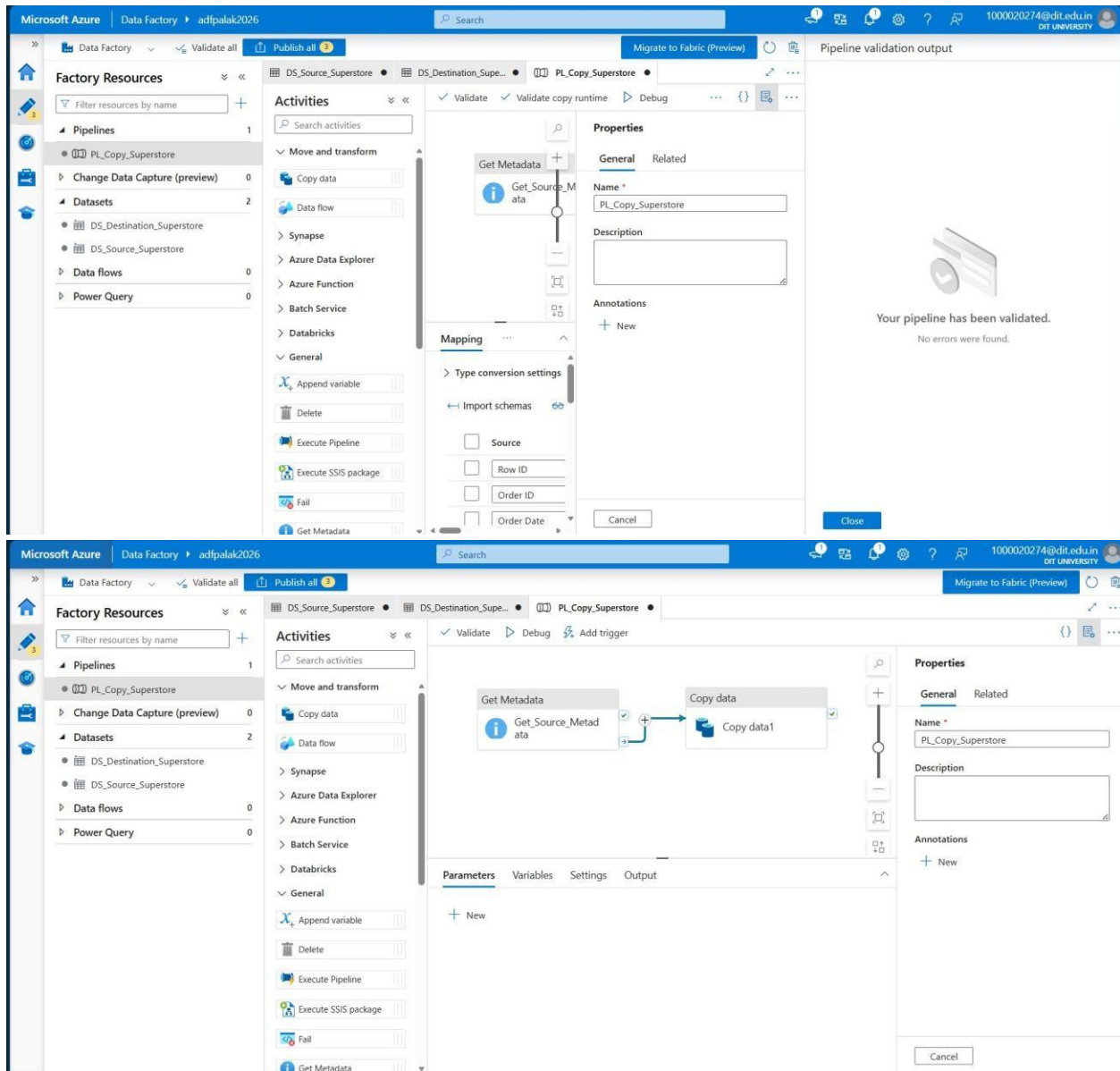
- Created a new pipeline named **PL_Copy_Superstore**.
- Added the **Get Metadata** activity to validate the source CSV file.
- Configured the **Copy Data** activity.
- Connected the Source Dataset with the Destination Dataset.
- Validated the pipeline before execution.

Result

The pipeline was successfully designed and configured.

Screenshots

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Task 5: Pipeline Execution

Objective

Execute the Azure Data Factory pipeline and verify successful execution.

Steps Performed

- Executed the pipeline using **Debug**.
- Published all pipeline changes.
- Triggered the pipeline manually.
- Opened the Monitor section to verify execution.

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- Confirmed that the pipeline completed successfully without any errors.
- Verified that the output CSV file was copied to the destination Blob container.

Result

The pipeline executed successfully and copied the data from the source container to the destination container.

Screenshots

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Pipeline validation output



Your pipeline has been validated.

No errors were found.

Close

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The screenshot shows the Azure Data Factory pipeline editor for a pipeline named 'PL_Copy_Superstore'. The pipeline consists of two activities: 'Get Metadata' (Get_Source_Metad ata) and 'Copy data' (Copy data1). The pipeline status is 'Succeeded'. The 'Output' tab is selected, showing the pipeline run ID '54f59c4b-0289-4d41-bcab-a996' and 'af6a4336'. The pipeline status is 'Succeeded'. The 'Output' tab shows the following table:

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime
Copy data1	✓ Succeeded	Copy data	7/11/2026, 10:13:36 PM	15s	AutoResolveIntegrationRuntime (Eas
Get_Source_Metad ata	✓ Succeeded	Get Metadata	7/11/2026, 10:13:30 PM	5s	AutoResolveIntegrationRuntime (Eas

The screenshot shows the Azure Data Factory Pipeline runs page for the pipeline 'PL_Copy_Superstore'. The page displays a list of pipeline runs with the following columns: Pipeline name, Run start, Run end, Duration, Triggered by, Status, Run, and Parameters. The table shows one run that succeeded.

Pipeline name	Run start	Run end	Duration	Triggered by	Status	Run	Parameters
PL_Copy_Superstore	7/11/2026, 10:18:21 PM	7/11/2026, 10:18:51 PM	30s	Manual trigger	✓ Succeeded	Original	

Task 6: IAM Role Assignment

Objective

Grant Azure Data Factory permission to securely access Azure Blob Storage using Azure Role-Based Access Control (RBAC).

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Steps Performed

- Opened the Azure Storage Account.
- Navigated to **Access Control (IAM)**.
- Assigned the **Reader** role to my Azure account.
- Assigned the **Storage Blob Data Contributor** role to the Azure Data Factory Managed Identity (**adfpalak2026**).
- Verified that the required permissions were successfully assigned.

Result

Azure Data Factory was granted permission to access Azure Blob Storage securely. The Managed Identity can now read and write files between Blob containers.

Mini Project: End-to-End Data Pipeline

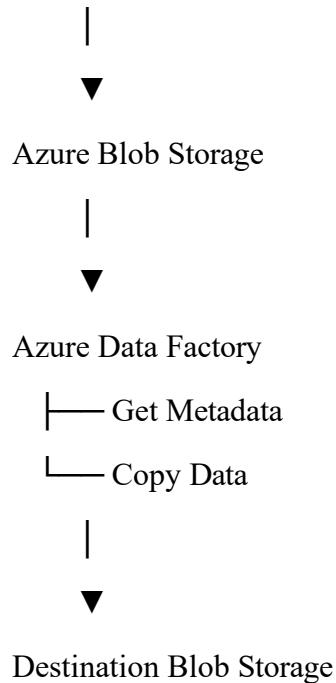
Problem Statement

Build a complete Azure Data Factory pipeline that reads a CSV file from Azure Blob Storage, validates the file using the Get Metadata activity, and copies the file to another Blob Storage location.

Pipeline Architecture

Sample-Superstore.csv

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Workflow

1. Uploaded the Superstore dataset into Azure Blob Storage.
2. Created Linked Services connecting Azure Data Factory to Azure Storage.
3. Created Source and Destination datasets.
4. Added the Get Metadata activity to validate the source file.
5. Configured the Copy Data activity.
6. Executed the pipeline using Debug and Trigger.
7. Monitored pipeline execution through Azure Data Factory Monitor.
8. Verified that the output file was copied successfully.
9. Configured IAM roles for secure access.

Final Output

- Source CSV uploaded successfully.
- Metadata validated successfully.
- Pipeline executed successfully.
- Data copied successfully to the destination Blob container.
- IAM permissions configured successfully.

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- End-to-end Azure Data Factory pipeline implemented successfully.

Learning Outcomes

Through this assignment, I gained practical experience with Microsoft Azure services and Azure Data Factory. I learned how to create and manage Azure resources, configure Azure Storage Accounts, create Blob containers, upload datasets, build Linked Services and Datasets in Azure Data Factory, design data pipelines using Get Metadata and Copy Data activities, execute and monitor pipelines, assign IAM roles using Azure RBAC, and verify successful data movement between Blob Storage containers. This assignment provided a clear understanding of building secure and scalable cloud-based data integration pipelines.

Conclusion

This assignment successfully demonstrated the implementation of an end-to-end Azure Data Factory pipeline. The project covered Azure resource creation, Blob Storage configuration, Azure Data Factory pipeline development, metadata validation, secure access management using IAM roles, and successful data transfer between Blob Storage containers. The objectives of the assignment were achieved successfully, providing hands-on experience with Azure cloud services and modern data engineering workflows.